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MNT/003 – Labelling of Operational Assets

1 Purpose

This document describes UK Distribution policy for the provision of statutory signs and notices on ground mounted substations, street furniture and on overhead line supports. These signs are intended to impart basic ownership details and safety information to members of the public and other third parties.

2 Scope

This document details UK Distribution's legal requirements for signs and notices that shall be fixed on ground mounted substations, street furniture and overhead line supports for the purpose of providing basic ownership details and safety information to members of the public and other third parties. It does not cover the labelling of plant and equipment for operational purposes, such as circuit nameplates on switchgear, overhead line support numbering etc.

3 Policy

3.1 Assessment of Relevant Drivers

Mid American has made a commitment within its corporate goal statements to be a leading safety performer within our industry sector. In addition, this policy will make a positive contribution to the following business values:

- Safety
- Regulatory, Political and Legal

3.1.1 Electricity Safety, Quality and Continuity, ESQC, Regulations (2002)

The ESQC Regulations (2002) place obligations on distributors to display safety signs and other notices at substations and on overhead line supports. The subsequent subsections detail the specific regulations relevant to this document.

3.1.1.1 Regulation 3(3)

This regulation requires that steps are taken to make the public aware of the dangers associated with activities in the vicinity of overhead lines. The DTI guidance document provides further information with regards the application of the regulation to specific sections of overhead line. The document advises that additional steps should be taken to mitigate the risk at sites that represent a significant risk of danger to the public, i.e. high-risk sites.

Although the regulations make no specific requirement to display signs at high-risk sites, the accompanying guidance document indicates that additional signs would be the minimum required to comply with the regulations.

3.1.1.2 Regulation 11(c)

This regulation requires that all distributors provide at "...every substation which he owns or operates...", the following signs;

- Sufficient safety signs to give warning of the dangers. The detailed design of the sign is detailed in Schedule 1 of the regulations.
- A notice giving the location or identification of the substation, the name of the distributor and "the telephone number where a person appointed by the distributor will be in constant attendance"
- Such other signs as may be necessary to give warning of danger.

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3.1.1.3 Regulation 19(2)

This regulation requires that sufficient safety signs are attached to all overhead line supports carrying high voltage conductors, or bare low voltage phase conductors, to provide warning of the dangers.

3.1.2 Control of Asbestos at Work Regulations (2002)

The Control of Asbestos at Work Regulations (2002) places obligations on the company to assess all work sites for the presence of asbestos. It places further obligations for sites where asbestos is, or is assumed to be, present. One of these further obligations is to provide information about the condition and location of the asbestos is provided to persons liable to disturb it and to label asbestos and materials containing asbestos in accordance with schedule 2 of those regulations.

In order to comply fully with the above regulations UK Distribution require a number of policies governing the management of assets and sites containing asbestos. This document is not intended to cover all obligations arising from the regulations. It only covers the requirements to provide information and to apply labels.

3.1.3 Health and Safety (Safety Signs and Signals) Regulations (1996)

The Health and Safety (Safety Signs and Signals) Regulations (1996) ensure that all signs erected to convey information regarding safety comply with specified colouring, wording and construction requirements. All signs and labels specified in this policy document shall comply with the requirements of these regulations.

3.2 Key Policy Requirements

The objectives of this policy are:

- To ensure that UK Distribution complies with all legislation relevant to the labelling of operational assets.
- To specify the requirements for the application of similar labelling to other operational assets not specifically included in the legislation.

It is not the objective of this policy to impose retrospective application of the requirements of this policy if there are no legal / statutory requirements to do so. If any existing labels from the legacy companies of YEDL and NEDL comply with the general “spirit” of this guidance then no further action need be taken. However, if there are legal / statutory requirements for particular signage, then existing signs that do not comply shall be changed to comply with those requirements.

3.2.1 Substations

All signs and notices shall be fitted such that they cannot provide any assistance to climbing.

The following shall be prominently displayed on, or adjacent to, all gates or doors giving access to substations.

- Statutory Safety Sign
- Property Notice giving details of ownership and emergency contact telephone number
- Nameplate with substation name and number

Where there is a site boundary, the following shall be prominently displayed on all doors and gates forming part of the boundary;

- Property Notice giving details of ownership and emergency contact telephone number

In addition:

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- Danger notices shall be displayed within substation boundaries such that they are visible from all reasonable angles of approach.

An exception to this is where the site is small and the Property Notice on the substation building door is clearly visible from the boundary. Under these circumstances it is not necessary to fit additional boundary labels.

Some distribution substations have a LV pillar which can be accessed without having to enter the substation itself. These will be treated as street furniture for the purpose of this document (See Section 3.2.2).

3.2.2 Street Furniture

The following shall be prominently displayed on the front access doors of all LV pillars (including haldo street lighting pillars and EHV cable oil pillars):

- Property Notice giving details of ownership and emergency contact telephone number
- Statutory Safety Sign

The covers of LV link box pits shall have no labels or signs fitted.

3.2.3 Labelling of Operational Assets for the presence of Asbestos

This section describes the policy for the labelling of Asbestos in its operational assets. These signs are intended to indicate the presence of Asbestos within an operational site to members of staff, public and other third parties who enter the area.

3.2.3.1 Labelling of Substations

3.2.3.1.1 Indoor Substations

If a survey has indicated that asbestos is found within the site, an **Asbestos Warning** label shall be affixed to the internal side of the outer most substation door. If the substation has multiple points of access additional labels shall be affixed to the internal side of every external door that can be used to gain access to the substation.

The label shall comply with Schedule 2 of the Control of Asbestos at Work regulations 2002. The label should be affixed using self-adhesive pads. A single laminated copy of the Asbestos Survey Form should accompany the label at the main point of entry.

If a survey has indicated that no asbestos is found within the site, an **Asbestos Checked** label shall be affixed to the internal side of the outer most substation door. Again additional labels shall be affixed to the internal side of every external door that can be used to gain access to the substation.

The label shall be 50mm in length and 25mm in height. The right hand side of the label shall carry the wording 'ASBESTOS CHECKED'. The lettering shall be white on a green background. The left hand side of the label shall have a white 'tick mark'. The label should be affixed using self-adhesive pads. A copy of the Asbestos Survey Form is not required to be left on site.

If subsequently any work is carried out that changes the condition of asbestos within the substation, then a laminated copy of the Asbestos Survey Form Update Sheet shall be left with the original Survey Form.

3.2.3.1.2 Outdoor Substations

If a survey has indicated that asbestos is found within the site, then a rigid **Asbestos Warning** label, along with a laminated copy of the Asbestos Survey Form shall be affixed to the switchgear in a zip tie bag. The label shall comply with Schedule 2 of the Control of Asbestos at Work regulations 2002.

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If a survey has indicated that no asbestos is found within the site, then a rigid **Asbestos Checked** label shall be to the switchgear in a zip tie bag. The label shall be 50mm in length and 25mm in height. The right hand side of the label shall carry the wording 'ASBESTOS CHECKED'. The lettering shall be white on a green background. The left hand side of the label shall have a white 'tick mark'. A copy of the Asbestos Survey Form is not required to be left on site.

If subsequently any work is carried out that changes the condition of asbestos within the substation, then a laminated copy of the Asbestos Survey Form Update Sheet shall be left with the original Survey Form.

3.2.3.1.3 Live Compounds

If a survey has indicated that asbestos is found within the live compound, then a self-adhesive **Asbestos Warning** label shall be affixed to the internal side of the fence post nearest the access point. The label shall comply with Schedule 2 of the Control of Asbestos at Work regulations 2002.

If during surveying no asbestos is found within the live compound, then no label is required within the live compound.

3.2.3.1.4 Oil Containing PCB's

Ground mounted transformers in service should be labelled in the following manner:

PCB below 50ppm Green label with white lettering

PCB contaminated 50 – 499ppm Yellow label with black lettering

PCB contaminated >500ppm Red label with white lettering

Labels should be 100mm x 150mm and clearly visible from all access doors / gates.

3.2.4 Overhead Lines

It is a legal requirement of the ESQC Regulations to fit Statutory Safety Signs to all supports carrying high voltage lines and low voltage bare conductor lines. Within UK Distribution Statutory Safety Signs will be fitted to all supports carrying lines regardless of voltage or conductor type.

In addition the regulations require that steps are taken to make the public aware of the dangers associated with activities in the vicinity of overhead lines. The DTI guidance document provides further information with regards the application of the regulation to specific sections of overhead line. The document advises that additional steps should be taken to mitigate the risk at sites, which may represent a significant risk of danger to the public. The UKD policy document *DSS/023 – Risk Assessment of Overhead Lines* details the sites at which additional safety signs should be applied. Sections 3.2.4.1.3 and 3.2.4.2.2 of this document provide details of the signs to be fitted at those sites.

The regulations also require that the degree of risk of danger from interference, vandalism or unauthorised access is assessed. The procedure for carrying out these assessments is contained within the UKD policy document *DSS/023 – Risk Assessment of Overhead Lines*. This document classifies some sites as 'Inadvertent Contact Sites' due to their proximity to the public and the risk of a member of the public making contact with the overhead line. These sites shall have signs fitted in accordance with sections 3.2.4.1.4 and 3.2.4.2.3 of this document (where consent can be gained from the land owner). These requirements represent the minimum standard to comply with the regulations, further mitigating actions may be undertaken dependant on the degree of risk assessed at that site. The need for further actions will be identified from the data collected during the high-risk site inspection, which is detailed within the UKD policy document *DSS/023 – Risk Assessment of Overhead Lines*.

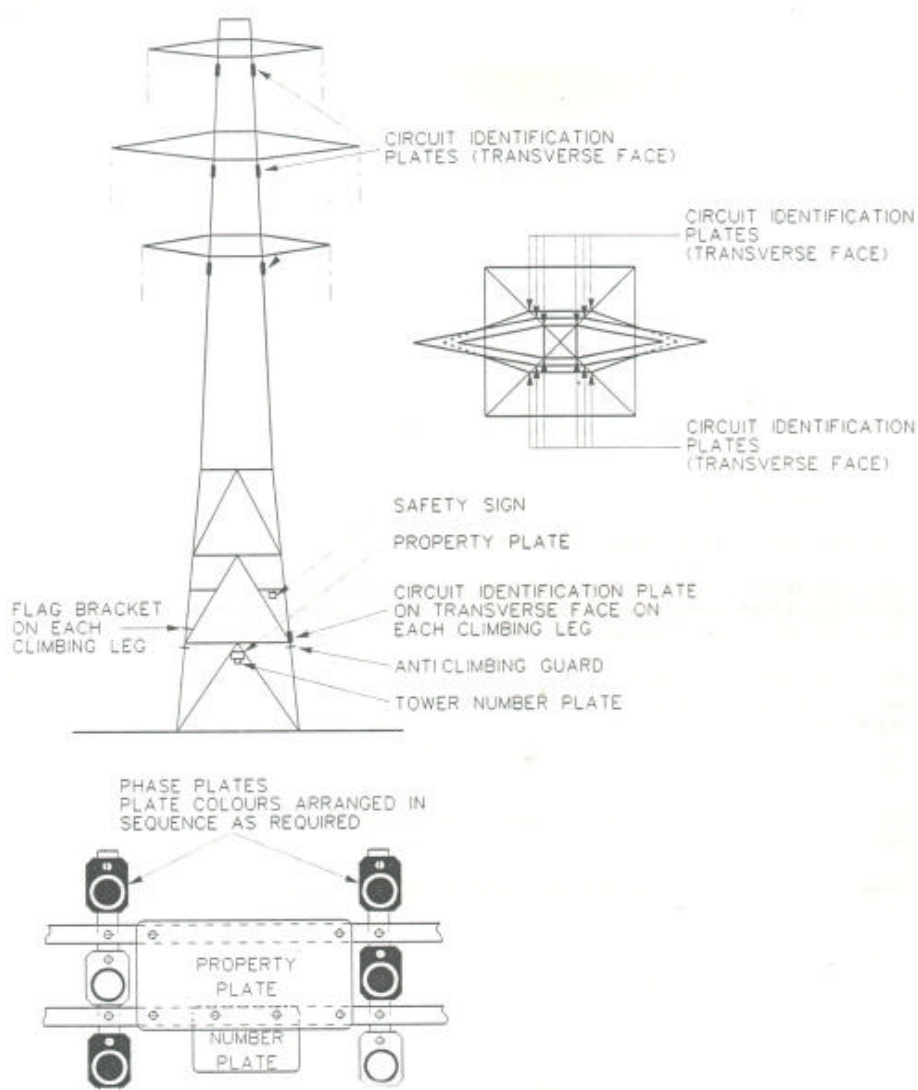
3.2.4.1 132kV Overhead Lines

In addition to the requirements of the ESQC Regulations property plates shall also be fitted to all 132kV overhead lines. Other labelling such as number plates, circuit id plates and phase colour plates shall also be fixed to all UK

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Distribution 132kV tower lines. The signs and labels shall be fitted in accordance with the diagram overleaf, safety signs and property plates are discussed in the following sections 3.2.4.1.1 and 3.2.4.1.2.



3.2.4.1.1 Statutory Safety Signs

Statutory safety signs shall be in accordance with Schedule 1 of 'The Electricity Safety, Quality and Continuity Regulations 2002'. They shall be positioned at a minimum height of 3 metres from ground level (or from where a person can stand). The exact position needs to be carefully considered to make sure that the sign is always displayed below the area of danger (bare connections etc). This is particularly important on towers supporting low level sealing end platforms.

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A minimum of two statutory safety signs shall be fitted to every tower on opposite transverse faces, i.e. facing along the route of the line. The shall be fixed on the first horizontal cross bracing above the anti-climbing guard and located close to the step bolt on the leg. If a sign is not normally visible from the direction of access to the tower then additional sign(s) should be fitted in a suitable position.

3.2.4.1.2 Property Plates

A property plate giving details of ownership and emergency contact telephone number shall be fitted on each tower. The plate shall be 610mm in length and 305mm in height. The lettering shall be black on a white background in accordance with BS 5499-5:2002, 'Graphical symbols and signs. Safety signs, including fire safety signs. Signs with specific safety meanings'.

3.2.4.1.3 Additional Safety Signs

For sites with towers requiring additional safety signs one safety sign shall be attached to each face of the tower fixed to the first horizontal bracing above the anti-climbing guard and located close to the step bolt on each leg.

For sites with other types of supports requiring additional safety signs three or four statutory safety signs shall be fitted, so that the signs make a continuous ring round the support. On portal support safety signs are needed on each limb of the structure.

The safety signs shall be in accordance with schedule 1 of the ESQC regulations 2002.

3.2.4.1.4 Inadvertent Contact Signs

One type of site requiring inadvertent contact signs is where the overhead line oversails, or is in the proximity of, fishing ponds. In these areas it is the intention that the existing signage, as obtained from the Electricity Association, will be used. At all other inadvertent contact sites the signs detailed in the following subsections will be the standard arrangement for fitting of warning and prohibition signs.

Warning Signs

A warning sign shall be placed on each access road to the site. The sign will carry the following language "Danger – Be aware - Look out, Look Up for Overhead Electric Power Lines"

The sign shall carry black lettering on a yellow background. There will be a danger of death symbol in the top left-hand corner as indicated on the diagram below. The sign will be 450mm in length and 300mm in height.



This sign will be placed at all entrances to the site, it will be fitted to two wooden posts and the posts will be concreted into the ground.

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Prohibition Signs

Prohibition signs shall be fitted adjacent to the overhead line. The sign will carry the following language “Danger – Overhead Electric Power Lines – Please be aware”.

The sign shall carry white lettering on a red background with a white outer border. There will be a danger of death symbol in the top left hand corner and a “prohibition” sign in the top right hand corner as indicated in the diagram below. The sign will be 450mm in length and 300mm in height.

This sign will be placed at a position 30metres either side of the overhead line, they will be fitted every 50metres along the length of the line and/or specifically where roads or access routes pass under the line. The prohibition signs will be fitted to two wooden posts and the posts will be concreted into the ground.



3.2.4.2 All Other Overhead Lines

Although not a legal requirement of the ESQC Regulations, other labelling such as number plates, circuit id plates and phase id labels, joint user support labels and railway crossing notices shall also be fixed to all UK Distribution EHV, HV and LV overhead lines.

Normally the signs and labels are placed as shown on the appropriate support arrangement drawings, facing along the direction of the line. The safety sign is always positioned highest with the appropriate additional labels grouped closely below it.

3.2.4.2.1 Statutory Safety Signs

Statutory safety signs shall be in accordance with Schedule 1 of ‘The Electricity Safety, Quality and Continuity Regulations 2002’. They shall be positioned at a minimum height of 3 metres from ground level (or from where a person can stand). The exact position needs to be carefully considered to ensure that the sign is always displayed below the area of danger (bare connections etc). This is particularly important on towers supporting low level sealing end platforms.

At least one safety sign shall be attached to every support of an overhead line. On portal supports safety signs are required on each limb of the structure. If a sign is not normal visible from the direction of access to the support then additional sign(s) shall be attached in suitable position(s).

On each EHV and HV support fitted with an anti-climbing guard two safety signs shall be fitted. The signs shall be positioned at the same level and on opposite faces of the support. At least one sign shall be positioned so as to be visible from the direction of access to the support. On portal supports signs are required on each leg of the structure.

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3.2.4.2.2 Additional Safety Signs

For sites with towers requiring additional safety signs one safety sign shall be attached to each face of the tower fixed to the first horizontal bracing above the anti-climbing guard and located close to the step bolt on each leg.

For sites with other types of supports requiring additional safety signs three or four statutory safety signs shall be fitted, so that the signs make a continuous ring round the support. On portal support safety signs are needed on each limb of the structure.

The safety signs shall be in accordance with schedule 1 of the ESQC regulations 2002.

3.2.4.2.3 Inadvertent Contact Signs

One type of site requiring inadvertent contact signs is where the overhead line oversails, or is in the proximity of, fishing ponds. In these areas it is the intention that the existing signage, as obtained from the Electricity Association, will be used. At all other inadvertent contact sites the signs detailed in the following subsections will be the standard arrangement for fitting of warning and prohibition signs.

Warning Signs

A warning sign shall be placed on each access road to the site. The sign will carry the following language “Danger – Be aware - Look out, Look Up for Overhead Electric Power Lines”

The sign shall carry black lettering on a yellow background. There will be a danger of death symbol in the top left-hand corner as indicated on the diagram below. The sign will be 450mm in length and 300mm in height.



This sign will be placed at all entrances to the site, it will be fitted to two wooden posts and the posts will be concreted into the ground.

Prohibition Signs

Prohibition signs shall be fitted adjacent to the overhead line. The sign will carry the following language “Danger – Overhead Electric Power Lines – Please be aware”.

The sign shall carry white lettering on a red background with a white outer border. There will be a danger of death symbol in the top left hand corner and a “prohibition” sign in the top right hand corner as indicated in the diagram below. The sign will be 450mm in length and 300mm in height.

This sign will be placed at a position 30metres either side of the overhead line, they will be fitted every 50metres along the length of the line and/or specifically where roads or access routes pass under the line. The prohibition signs will be fitted to two wooden posts and the posts will be concreted into the ground.

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3.3 Implementation and Monitoring of this Policy

Several departments within UK Distribution are accountable for the implementation and monitoring of the labelling of operational assets policy in accordance with the following table:

Designation	Role
Construction Manager	Responsible for the implementation of all elements of the labelling of operational assets policy relevant to construction activities and will ensure responsible persons are appointed to implement this policy.
Maintenance Manager	Responsible for the implementation of all elements of the labelling of operational assets policy relevant to maintenance activities and will ensure responsible persons are appointed to implement this policy.
Safety Manager	Monitor compliance of those accountable for implementation of the labelling of operational assets policy.

3.4 Planned Policy Review

This policy shall be proposed for review on a biennial basis or at any time when external or internal influences drive a change in policy e.g. a change in legislation.

The following responsibilities shall apply to policy control and review:

Designation	Responsibility
Publication Manager	Responsible for issuing a quarterly report to the Policy Production Manager (or representative) detailing policies scheduled for biennial review within the next 6 months
Policy Production Manager	Responsible for assessing the continued applicability of this company policy and for amending this document and communicating any changes in policy.

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4 References

4.1 External Documentation

Reference	Title	Version and date
Statutory Instrument 1996 No. 341	Health and Safety (Safety Signs and Signals) Regulations 1996	18 th February 1996
Statutory Instrument 2002 No. 2665	The Electricity Safety, Quality and Continuity Regulations 2002	24 th October 2002
Statutory Instrument 2002 No. 2675	Control of Asbestos at Work Regulations 2002	24 th October 2002
British Standard 5499 Part 5.	BS 5499-5:2002, 'Graphical symbols and signs. Safety signs, including fire safety signs. Signs with specific safety meanings'.	2002

4.2 Internal documentation

Reference	Title	Version and date
DSS/023	<i>Risk Assessment of Overhead Lines</i>	V1 Oct 2003

5 Definitions

Term	Definition
LV Link Box	An enclosure installed in a 'pit' below ground level containing low voltage cable terminations and which may contain low voltage switchgear/fusegear. Access to the live equipment is normally gained by lifting a 'cover' installed in and flush with the pavement surface. A special tool is required for this operation.
LV Pillar	An enclosure above ground level containing low voltage cable terminations and which may contain low voltage switchgear/fusegear. Access to the live equipment is normally via a door in the front of the pillar. A special key is required to unlock these doors.
Nameplate Sign	A sign giving details of the asset name and any identification number. The dimensions of the label must be at least 150mm x 100mm, with a text size of 10mm or greater. An example of the sign is shown in Photo 3.
NEDL	Northern Electric Distribution Limited
Property Notice	A statutory sign giving the name of the Company and the address and telephone number where the Company may be contacted. The notice is shown in Photo 2
Safety Sign	A statutory sign giving warning of 'Danger of Death'. The letters and symbols are black on a yellow background. The sign is shown in Photo 1.
Secure Area	An area of the site from which it is intended to exclude all but persons authorised to gain access to the area.
Security Boundary	An unclimbable building, fence or wall enclosing a secure area.
Site Boundary	A fence or similar measure for the sole purpose of indicating the limits of Yorkshire Electricity's occupancy and responsibility at a distribution substation site.
Street Furniture	Low voltage plant and equipment which is generally located in or adjacent to public footpaths. It is accessible to members of the public going about their normal business. Also includes haldo street lighting pillars and EHV cable oil pillars.
UK Distribution	NEDL and YEDL
Unclimbable	Being very difficult to climb over without the use of aids or extreme determination.
YEDL	Yorkshire Electricity Distribution plc

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6 Authority for Issue

1. Policy Author

I sign to confirm that I have completed and checked this policy and I am satisfied with its content

		Sign	Date
Stephen Murray	Network Investment Engineer	Stephen Murray	10/10/03

2. Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this policy and submit it for approval and authorisation.

		Sign	Date
Mark Nicholson	Network Policy Manager	Mark Nicholson	10/10/03

3. Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this policy and submit it for approval and authorisation.

		Sign	Date
Mark Drye	Network Integrity Performance Island Leader	Mark Drye	10/10/03
Mick Hickling	Safety and Environment Performance Island Leader	Mick Hickling	10/10/03

4. DBD Assurance

I sign to confirm that this policy has been assured for issue on to the DBD system

		Sign	Date
Sean Johnson	Safety Engineer	Sean Johnson	10/10/03

5. Approval

Approval is given for the content of this policy

		Sign	Date
Phil Jones	Strategy and Investment Director	Phil Jones	10/10/03

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6. Authorisation

Authorisation is granted for publication of this policy

		Sign	Date
Peter McCormick	Utilities Safety, Environment & Training Director	Peter McCormick	10/10/03